Xiaojun Mei

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EDUCATION

- 2018 – 2021 Shanghai Maritime University (PhD)
  Major: Traffic information engineering and control
- 2019 – 2020 University of Lisbon (Visiting PhD student)
  Major: Electronic engineering
- 2016 – 2018 Shanghai Maritime University (Master)
  Major: Vehicle operation engineering
- 2015 – 2015 Mokpo Maritime University (Visiting B.E. student)
  Major: Navigation technology
- 2012 – 2016 Shanghai Maritime University (Bachelor)
  Major: Navigation technology

WORK EXPERIENCE

- 2023 – Present Associate Professor, Shanghai Maritime University
- 2026 – 2027 Visiting Researcher, KAUST
- 2021 – 2023 Postdoctoral Fellow, Shanghai Maritime University

RESEARCH INTERESTS

- Marine/Ocean Engineering
- Wireless Communications
- Intelligent Vessels and Ocean Monitoring Sensor Networks
- Maritime Communications, Target Localization, and Navigation

PROJECTS

PRINCIPLE INVESTIGATOR

- Research on Robust Localization and Tracking in Ocean Sensor Networks based on Quantum Optimization Framework, Grant No. 52201401, Ongoing (**Sponsored by Natural Science Foundation of China**)
- Research on Resource Localization Mechanism and Acquisition Strategy of Dynamic Uncertain Ocean Sensor Network, Grant No. 24CGA52, Ongoing (**Sponsored by Shanghai Education Development Foundation and Shanghai Municipal Education Commission**)
- Theory and Key Technologies for Optimal Dynamic Deployment of Polar Situational Awareness Cluster System (**Sub-Project PI**), Grant No. 52331012, Ongoing (**Sponsored by Natural Science Foundation of China**)
- Research on Three-Dimensional Perception for Maritime Transportation Based on Hybrid Acoustic–Optical Wireless Networks, Grant No. 202508310266, Ongoing (**Sponsored by China Scholarship Council**)
- Multi-Source Sea Ice Situation Fusion Perception and Dynamic Vessel Route Planning for Polar Navigation, Ongoing, (**Sponsored by National Key Laboratory of Waterway Traffic Control**)
- Research on Target Rescue Strategy of Mobile Ocean Sensor Networks under Uncertain Conditions, Grant No. 24-16901, Ongoing (**Sponsored by Shanghai Maritime University**)
- Research on Target Localization and Tracking in Underwater Wireless Sensor Networks with Quantum Coupling Optimization Framework, Grant No. 2022M712027, Completed (**Sponsored by Postdoctoral Science Foundation of China**)
- Key Technology for Target Localization in High Accuracy and High Reliability Ocean Monitoring Sensor Network, Grant No. 2022767, Completed (**Sponsored by Shanghai Human Resources and Social Security Bureau**)
- Applications of Wireless Sensor Networks on Water Transportation, Grant No. 201908310079, Completed (**Sponsored by China Scholarship Council**)
- Research on Localization in Ocean Sensor Networks in the presence of Uncertainty, Grant No. 2019YBR002, Completed (**Sponsored by Shanghai Maritime University**)
- Research on Cooperative Localization Algorithms Based on Modified Particle Filter in Marine Monitoring Wireless Sensor Networks, Grant No. 2017ycx030, Completed (**Sponsored by Shanghai Maritime University**)

PARTICIPATION

- Research on Key Technologies and System Development of Communication and Navigation Support in the Arctic Waterway, Grant No.2021YFC2801002, Completed (**Sponsored by National Ministry of Science and Technology**)
- Research on Key Technologies for Robust and Efficient Data Transmission in Ocean Three-Dimensional Observation Networks, Grant No. 23CGA61, Completed (**Sponsored by Shanghai Education Development Foundation and Shanghai Municipal Education Commission**)

- Dynamic Self-adaptive Clustering Based Intelligent Data Prediction and Reconstruction in Ocean Sensor Networks, Grant No. 52071200 (**Sponsored by Natural Science Foundation of China**)
- Three-Dimensional Dynamic Cooperative Localization Mechanism of Marine Sensor Networks Based on Wave Shadowing Effect Model, Grant No. 51579143 (**Sponsored by Natural Science Foundation of China**)
- Coastal Meteorological Monitoring and Warning System Based on Buoy Internet of Things and Its Navigation Aid Application, Grant No. 18040501700 (**Sponsored by Shanghai Science and Technology Committee**)
- Key technical research on positioning based wireless sensor networks and target search and rescue at sea, Grant No. 12SG40 (**Sponsored by Shanghai Municipal Education Commission and Shanghai Education Development Foundation**)

PUBLICATION

PEER-REVIEWED JOURNAL ARTICLES

1. **X. Mei**, H. Wu, J. Xian, X. Chen, and Y. Zang. "Theoretical Analysis of Hybrid RSS and TOA Localization with Stratified Propagation under Inaccurate Anchors in UAIoTs", *Intelligent Marine Technology and Systems*, vol. 4, p. 2, 2026. (EI)
2. H. Wu, Y. Zhong, K. Zhang, L. Zhou, Y. Zhang, **X. Mei**, Q. Zhang, Y. Li, and T. Sun. "Improved A∗+ D∗ Lite Hierarchical Framework Path Planning Algorithm for Heterogeneous Sea Ice Scenarios", *Ocean Engineering*, vol. 343, p. 123387, 2026. (SCI, JCR Q1)
3. **X. Mei**, H. Wu, F, Miao, and D. Han. "Quantum-inspired Optimal Sensor Deployment for 3D Localization under Uncertain Target in Stratified Oceanic Environment". *Ocean Engineering*, vol. 341, pp. 122646, 2025. (SCI, JCR Q1)
4. J. Xian, J. Ma, H. Wu, **X. Mei**, X. Chen, Y. Zhang, W. Wang, and W. Li. "Novel robust and accurate localization algorithm for underwater search and rescue wireless sensor networks under unknown marine dynamic environmental factors", *Ocean Engineering*, vol. 339, p. 122107, 2025. (SCI, JCR Q1)
5. **X. Mei**, H. Wu, N. Saeed, D. Han, and K-C Li. "Stratification-Aware RSS-Based Localization in Underwater Environments: CRLB Analysis and Perturbation-Resilient Solutions", *IEEE Wireless Communications Letters*, vol. 11, no.10, p. 17757-17769,2025. (SCI, JCR Q1)
6. J. Zhao, Y. Peng, B. Han, **X. Mei**, H. Li, and Y. Liu. "TSeq-GAN: a generalized and robust blind source separation framework for AIS signals of unmanned surface vehicles", *Frontiers in Marine Science*, vol. 12, p. 1635614, 2025. (SCI, JCR Q1)
7. **X. Mei**, H. Wu, X. Chen, J. Xian, Z. Wu, "Three-Step Localization Algorithm in WSNs under Two Uncertain Parameters", *Journal of Electronic Measurement and Instrument*, vol. 39, no. 5, pp. 19-28, 2025 (EI)
8. X. Liu, L. Huang, **X. Mei***, N. Saeed, F. Wang, Y. Zhang, X. Ma, C. Weng, "Multi strategy fusion enhanced channel estimation algorithm based on deep learning", *Ain Shams Engineering Journal*, vol. 16, no. 7, pp. 103416, 2025. (SCI, JCR Q1)

9. J. Xian, J. Ma, X. Chen, H. Wu, **X. Mei**, Y. Zhang, W. Wang, "Robust and Precise Drowning Target Localization under Dual-parameters Uncertainty", *Regional Studies in Marine Science*, vol. 85, pp. 104131, 2025. (SCI, JCR Q2)
10. L. Liang; H. Wu; **X. Mei**; Q. Zhang; Y. Zhang; J. Xian, K-C Li, "Robust Target Localization in WSNs: A RotQCP Approach for NLOS Mitigation", *IEEE Internet of Things Journal*, vol. 12, no. 18, p. 36955-36965, 2025 (SCI, JCR Q1)
11. H. Zhang, G. Niu, T. Liu, C. Qian, W. Zhao, **X. Mei**, H. Wu, "Status Quo and Future Prospects of China's Weather Routing Services for Ocean-Going Business Vessels", *Oceans*, vol. 6, no. 3, pp. 38, 2025. (ESCI)
12. Y. Zhang, T. Gulliver, H. Wu, J. Li, **X. Mei**, J. Xian, K-C Li, "3-D RSSD Localization Under Mixed Gaussian Noise and NLOS Environments in UWSNs", *IEEE Internet of Things Journal*, vol. 12, no. 14, p. 28731- 28742, 2025. (SCI, JCR Q1)
13. H. Li, F. Miao, and **X. Mei**, "Facilitating Multi-UAVs application for rescue in complex 3D sea wind offshore environment: A scalable Multi-UAVs collaborative path planning method based on improved coatis optimization algorithm", *Ocean Engineering*, vol. 324, pp. 120701, 2025. (SCI, JCR Q1)
14. J. Xian, J. Ma, **X. Mei**, H. Wu, N. Saeed, D. Han, M. Marino, K-C Li, "Robust Coarse-to-Fine 3-D-Target-Localization Algorithm for Underwater-IoT-Based Networks: Design and Performance Evaluation Under Uncertain Multiparameters", *IEEE Internet of Things Journal*, vol. 12, no. 13, p. 25119- 25135, 2025 (SCI, JCR Q1)
15. W. Wang, H. Wu, S. Yang, **X. Mei**, D. Han, M. Marino, K-C Li, "LNPP: Logical Neural Path Planning of Mobile Beacon for Ocean Sensor Networks in Uncertain Environments using Hierarchical Reinforcement Learning", *IEEE Transactions on Network Science and Engineering*, vol. 12, no. 4, p. 2606-2621, 2025. (SCI, JCR Q1)
16. Y. Zhang, H. Wu, J. Xian, **X. Mei**, K. Zhang, Q. Zhang, F. Wang, "A Multi-head ProbSparse Self-attention Mechanism based High-Precision and High-Robustness Reconstruction Model for Missing Ocean Data", *IEEE Sensors Journal*, vol. 25, no. 8, p. 13374-13385, 2025. (SCI, JCR Q1)
17. Y. Zhang, J. Li, T. Gulliver, H. Wu, G. Xie, **X. Mei***, J. Xian, W. Wang, L. Liang, "Metaheuristic Optimization for Robust RSSD-Based UAV Localization with Position Uncertainty", *Drones*, vol. 9, no. 2, pp: 147, 2025. (SCI JCR Q1)
18. J. Xian, J. Ma, H. Wu, **X. Mei**, F. Tan, et al. "Coarse-to-Fine Localization Method for UASNs under Unknown Multi-Parameters", *Control and Decision*, vol. 40, no. 1, p. 38-47, 2025. (EI, CSCD)
19. J. Xian, Z. Li, H. Wu, W. Wang, X. Chen, **X. Mei**, Y. Zhang, B. Han, J. Ma, "Novel Polarization Construction Method and Synchronization Algorithm for Underwater Acoustic Channel Under T-Distribution Noise Environment", *Journal of Marine Science and Engineering*, vol. 13, no. 2, pp: 362, 2025. (SCI, JCR Q1)
20. H. Wu, K. Zhang, **X. Mei**, L. Liang, Z. Zhang, F. Wang, B. Han, D. Han, K-C Li, "Heuristic Strategy-Empowered Real-Time Path Following for Autonomous Surface Vessel With Adaptive Line-of-Sight Guidance", *IEEE Journal of Oceanic Engineering*, vol. 50, no.1, PP: 307-323, 2025. (SCI, JCR Q1)

21. Q. Zhang, H. Wu, L. Liang, **X. Mei**, J. Xian, "Robust sensor selection based on maximum correntropy criterion for ocean data reconstruction", *Frontiers in Marine Science*, vol. 11, pp: 1467519, 2024. (SCI, JCR Q1)

22. F. Miao, H. Li, and **X. Mei**. "Three-Dimensional Path Planning of UAVs for Offshore Rescue Based on a Modified Coati Optimization Algorithm", *Journal of Marine Science and Engineering*, vol. 12, no. 9, p:1676, 2024（SCI, JCR Q1）

23. Y. Zhang, T. Gulliver, H. Wu, **X. Mei**, J. Li, and F. Lu. "An Efficient Estimator for Source Localization in WSNs using RSSD and TDOA Measurements", *Pervasive and Mobile Computing*, vol. 102, p: 101936, 2024.（SCI, JCR Q2）

24. Q. Zhang, H. Wu, L. Liang, X. Mei, J. Xian, and Y. Zhang. "A Robust Sparse Sensor Placement Strategy Based on Indicators of Noise for Ocean Monitoring", *Journal of Marine Science and Engineering*, vol. 12, no. 7, p:1220, 2024（SCI, JCR Q1）

25. F. Miao, H. Li, G. Yan, **X. Mei***, Z. Wu, W. Zhao, T. Liu, and H. Zhang, "Optimizing UAV Path Planning in Maritime Emergency Transportation: A Novel Multi-strategy White Shark Optimizer", *Journal of Marine Science and Engineering*, vol. 12, no. 7, p.1207, 2024.（SCI，JCR Q1）

26. **X. Mei**, F. Miao, W. Wang, H. Wu, B. Han, Z. Wu, X. Chen, J. Xian, Y. Zhang, and Y. Zang, "Enhanced Target Localization in the Internet of Underwater Things through Quantum-Behaved Metaheuristic Optimization with Multi-Strategy Integration", *Journal of Marine Science and Engineering*, vol. 12, no. 6, p.1024, 2024.（SCI，JCR Q1）

27. **X. Mei**, D. Han, N. Saeed, H. Wu, B. Han, and K-C. Li, "Localization in Underwater Acoustic IoT Networks: Dealing with Perturbed Anchors and Stratification", *IEEE Internet of Things Journal*, vol. 11, no. 10, p. 17757-17769, 2024.（SCI，JCR Q1）

28. X. Chen, C. Wei, Y. Yang, L. Luo, S. A. Biancardo and **X. Mei**, "Personnel Trajectory Extraction From Port-Like Videos Under Varied Rainy Interferences", *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 6567-6579, 2024.（SCI, JCR Q1）

29. Q. Zhang, H. Wu, **X. Mei**, D. Han, M.D. Marino, and S. Guo. "A Sparse Sensor Placement Strategy based on Information Entropy and Data Reconstruction for Ocean Monitoring", *IEEE Internet of Things*, vol. 10, no. 22, p. 19681-19694, 2023. (SCI, JCR Q1)

30. S. Shi, B. Han, Z. Wu, D. Han, H. Wu, and X. Mei. "BLSAE-SNIDS: A Bi-LSTM Sparse Autoencoder Framework for Satellite Network Intrusion Detection", *Computer Science and Information Systems*, [Online] doi: 10.2298/CSIS240401041S, 2024（SCI, JCR Q3）

31. **X. Mei**, D. Han, N. Saeed, H. Wu, F. Miao, J. Xian, X. Chen, and B. Han, "Navigating the Depths: A Stratification-Aware Coarse-to-Fine Received Signal Strength-based Localization for Internet of Underwater Things", *Frontiers in Marine Science*, vol. 10, p. 1210519, 2023.（SCI, JCR Q1）

32. **X. Mei**, D. Han, Y. Chen, H. Wu, and T. Ma, "Target Localization using Information Fusion in WSNs-based Marine Search and Rescue", *Alexandria Engineering Journal*, vol. 68, p. 227-238, 2023. (SCI, JCR Q1)

33. J. Ma, J. Xian, H. Wu, Y. Yang, **X. Mei**, Y. Zhang, X. Chen, and C. Zhou. "Novel High-Precision and High-Robustness Localization Algorithm for Underwater-Environment-

Monitoring Wireless Sensor Networks", *Journal of Marine Science and Engineering*, vol. 11, no. 9, p. 1713, 2023. (SCI, JCR Q1)

34. Z. Sun, D. Han, D. Li, T.H. Weng, K.C. Li, and **X. Mei**. "MedRSS: A blockchain-based scheme for secure storage and sharing of medical records", *Computers & Industrial Engineering*, vol. 183, no. 1, p. 109521, 2023. (SCI, JCR Q1)
35. H. Wu, Y. Zhang, L. Liang, **X. Mei**, D. Han, B. Han, T.H. Weng, and K-C. Li. "Multi-head attention-based model for reconstructing continuous missing time series data", *Journal of Supercomputing*, vol. 2023, 2023. (SCI, JCR Q2, IF: 3.3)
36. H. Wu, F. Wang, **X. Mei**, L. Liang, B. Han, D. Han, T.H. Weng, and K-C. Li. "A novel fuzzy control path planning algorithm for intelligent ship based on scale factors", *Journal of Supercomputing*, vol. 2023, 2023. (SCI, JCR Q2)
37. **X. Mei**, D. Han, N. Saeed, H. Wu, T. Ma, and J. Xian, "Range Difference-based Target Localization under Stratification Effect and NLOS bias in UWSNs", *IEEE Wireless Communications Letters*, vol. 11, no. 10, p. 2080-2084, 2022. (SCI, JCR Q1)
38. **X. Mei**, D. Han, N. Saeed, H. Wu, C. Chang, B. Han, T. Ma, and, J, Xian, "Trajectory Optimization of Autonomous Surface Vehicles with Outliers for Underwater Target Localization", *Remote Sensing*, vol. 14, no. 17, p. 4343, 2022. (SCI, JCR Q1)
39. **X. Mei**, Y. Chen, X. Xu, and H. Wu, "RSS Localization using Multi-step Linearization in the presence of Unknown Path Loss Exponent", *IEEE Sensors Letters*, vol.6, no.8, p.1-4, 2022. (ESCI, EI)
40. J. Xian, H. Wu, **X. Mei**, Y. Zhang, X. Chen, Q. Zhang, and L. Liang. "Novel Energy-Efficient Opportunistic Routing Protocol for Marine Wireless Sensor Networks Based on Compressed Sensing and Power Control", *Journal of Ocean University of China*, vol. 21, no. 6, p. 1504-1516, 2022.（SCI，JCR Q4）
41. J. Xian, H. Wu, **X. Mei**, X. Chen, and Y. Yang, "Low-Delay and Energy-Efficient Opportunistic Routing for Maritime Search and Rescue Wireless Sensor Networks", *Remote Sensing*, vol. 14, no. 20, p. 5178, 2022.（SCI，JCR Q1）
42. H. Wu, Y. Hu, W. Wang, **X. Mei**, and J. Xian, "Ship Fire Detection Based on an Improved YOLO Algorithm with a Lightweight Convolutional Neural Network Model", *Sensors* vol. 22, no. 19, p. 7420, 2022.（SCI，JCR Q2）
43. T. Ma, W. Zhang, Y. Li, Y. Zhao, Q. Zhang, **X. Mei**, and J. Fan, "Communication-constrained cooperative bathymetric simultaneous localisation and mapping with efficient bathymetric data transmission method", *Journal of Navigation*, 1-17, 2022. Doi:10.1017/S0373463321000904.（SCI, JCR Q2）.
44. Y. Zhang, H. Wu, **X. Mei**, et al, "Unknown Transmit Power RSSD-based Localization under Gaussian Mixture Channel", *IEEE Sensors Journal,* vol. 22 , no.9, pp.9114-9123, 2022 (SCI, JCR Q1).
45. H. Wu, L. Liang, **X. Mei,** and Y. Zhang, "A Convex Optimization Approach for NLOS Error Mitigation in TOA-based Localization", *IEEE Signal Processing Letters*, vol. 29 pp.677-681, 2022（SCI, JCR Q2）.

46. **X. Mei**, H. Wu, J. Xian, and T. Ma, "Information-driven Optimal Placement for Target Localization in Ocean Sensor Networks", *Journal of Huazhong University of Science and Technology (Natural Science Edition)*, vol. 49, no.11, p.23-29, 2021 (EI)
47. **X. Mei**, H. Wu, J. Xian, and B. Chen, "RSS-based Byzantine Fault-tolerant Localization Algorithm under NLOS Environment," *IEEE Communications Letters*, vol. 25, no.2, p.474-478, 2021. (SCI, JCR Q2)
48. Y. Zhang, H. Wu, **X.Mei**, J. Xian, W. Wang, Q. Zhang, and L. Liang, "Two-Phase Robust Target Localization in Ocean Sensor Networks using Received Signal Strength Measurements," *Sensors*, vol. 21, no.5, p.1724, 2021. (SCI, JCR Q2)
49. S. Cheng, H. Wu, and **X. Mei**, "An alternative nonnegative constrained framework-based cooperative localization algorithm in ocean sensor networks," *Computer Engineering Application*, vol. 57, no. 23, pp. 129–136, 2021. (EI)
50. **X. Mei**, H. Wu, and J. Xian, "Matrix Factorization based Target Localization via Range Measurements with Uncertainty in Transmit Power," *IEEE Wireless Communications Letters*, vol. 9, no. 10, p.1611-1615, 2020. (SCI, JCR Q1)
51. **X. Mei**, H. Wu, N. Saeed, T. Ma, J. Xian, and Y. Chen, "An Absorption Mitigation Technique for Received-Signal-Strength-Based Target localization in Underwater Wireless Sensor Networks", *Sensors*, vol. 20, no. 17, p. 4698, 2020. (SCI, JCR Q2)
52. J. Xian, H. Wu, **X. Mei**, Y. Zhang, H. Chen, and J. Wang, "NMTLAT: A New robust mobile Multi-Target Localization and Tracking Scheme in marine search and rescue wireless sensor networks under Byzantine attack," *Computer Communications*, vol. 160, pp. 623–635, 2020. (SCI, JCR Q1)
53. Y. Chen, Y. Hu, S. Zhang, **X. Mei**, and Q. Shi, "Optimized Erosion Prediction with MAGA Algorithm Based on BP Neural Network for Submerged Low-Pressure Water Jet," *Applied Sciences*, vol. 10, no. 8, p. 2926, Apr. 2020. (SCI, JCR Q3)
54. **X. Mei**, H. Wu, J. Xian, B. Chen, H. Zhang, and X. Liu, "A Robust, Non-Cooperative Localization Algorithm in the Presence of Outlier Measurements in Ocean Sensor Networks," *Sensors*, vol. 19, no. 12, p. 2708, 2019. (SCI, JCR Q2)
55. H. Wu, J. Xian, **X. Mei**, Y. Zhang, J. Wang, J. Cao, and P. Mohapatra, "Efficient target detection in maritime search and rescue wireless sensor network using data fusion," *Computer Communnications*, vol. 136, pp. 53–62, 2019. (SCI, JCR Q1)
56. H. Wu, Q. Meng, J. Xian, **X. Mei**, C. Claramunt, and J. Cao, "An Information Entropy Based Event Boundary Detection Algorithm in Wireless Sensor Networks," *Symmetry (Basel).*, vol. 11, no. 4, p. 537, Apr. 2019. (SCI, JCR Q2)
57. Y. Zhang, H. Wu, J. Xian, and **X. Mei**, "Adaptive Clustering Algorithm in OceanWireless Sensor Network Under Double Constraints," *Computer Engineering Application*, vol. 19, no. 55, pp. 128–133, 2019. (EI)
58. H. Wu, **X. Mei**, X. Chen, J. Li, J. Wang, and P. Mohapatra, "A novel cooperative localization algorithm using enhanced particle filter technique in maritime search and rescue wireless sensor network.," *ISA Transactions*, vol 78, p.39-46, 2018. (SCI, JCR Q1)

59. **X. Mei**, H. Wu, Y. Chen, and E. Jiang, "Ship tracking of wireless sensor network based on improved adaptive particle filter," *Journal of Shanghai Maritime Univiversity*, vol. 39, no. 2, pp. 12–16, 2018.

MANUSCRIPTS IN PREPARATION/SUBMITTED FOR REVIEW

1. **X. Mei**, et. al, "Fundamental Limits of RSS-based Localization in Underwater Acoustic IoT Networks in the presence of Multiple Perturbed Parameters", (In preparation and planning to submit to IEEE Transactions on xxxx)
2. **X. Mei**, et. al, "IMM-based Underwater Target Tracking in the presence of Stratification Effect", (In preparation and planning to submit to JMSE)
3. **X. Mei**, et. al, "Metaheuristic Algorithms-based Target Localization in the Internet of Underwater Things: Simulated Experiments and Comparisons", (Invited paper and planning to submit to Electronics and signal processing)

CONFERENCE AND PATENT

CONFERENCE

1. W. Yuan, **X. Mei***. "Brief Literature Survey on Localization in Ocean Wireless Sensor Networks", *Journal of Physics: Conference Series*, vol. 3055, no. (1), 012010, 2025.
2. C. Chen, **X. Mei***. "A Review of Ranging Optimization Algorithms for Indoor Positioning", *Journal of Physics: Conference Series*, vol. 3055, no. (1), 012045, 2025.
3. B. Liu, H. Wu, **X. Mei**, Y. Zhang "Polar Dense Ice Layer Ship Path Planning Based on DI-IVYA-A* Algorithm", *the 6th International Conference on Computer Engineering and Application (ICCEA 2025), Hangzhou, China, 2025*.
4. B. Sun, **X. Mei***, Y. Du, M. Gao, "A General Intelligence Theory Based on the Representation of Need," *the 9th International Conference on Computer and Communication Systems (ICCCS 2024), Xi'an, China, 2024*.
5. H. Wu, X. Zhao, **X. Mei***, B. Han, and Z. Wu, "An RSSI-Based Fingerprint Localization using Multi-Signal Mean Optimization Filter in Indoor Environment Onboard a Passenger Ship," *the 9th International Conference on Computer and Communication Systems (ICCCS 2024), Xi'an, China, 2024.*
6. H. Wu, C. Jiao, **X. Mei***, B. Han, Z. Wu, Y. Zhang, "Fast Bidirectional A-Star Algorithm Based on Radar Image Data Fusion for Polar Ship Route Planning," *the 9th International Conference on Computer and Communication Systems (ICCCS 2024), Xi'an, China, 2024*.
7. **X. Mei**, H. Wu, D. Han, X. Chen, J. Xian, and B. Han, "A Computationally Efficient Target Localization Algorithm in Underwater Wireless Sensor Networks," *the 8th International Conference on Computer and Communication Systems (ICCCS 2023), Guangzhou, China, 2023*.
8. **X. Mei**, D. Han, Y. Chen, H. Wu, and J. Xian, "An Actively Positioning System with an Accurate Localization Technique for Sunken Container via a Three-layer Network Structure," *Asia Navigation Conference, Toyama, Japan, 2022*.

9. **X. Mei**, H. Wu, Y. Chen, H. Zhang, and Q. Zhang, "A Lightweight Computation Target Localization Algorithm using Information Fusion in WSNs-based Marine Search and Rescue," *Asia Navigation Conference, Tianjin, China, 2021*
10. **X. Mei**, H. Wu, J. Xian, H. Zhang, and Y. Zhang, "A Robust Localization with Outlier Measurements in Underwater Sensor Networks", *Proceedings of the 2019 Academic Conference of the Chinese Acoustics Society Hydroacoustic Branch, Nanjing, China, 2019*.
11. X. Liu, R. Han, **X. Mei**, and Y. Hu. "A novel UAV target detection method based on the improved YOLOv4-tiny network", *2023 3rd International Conference on Computer Vision and Pattern Analysis (ICCPA 2023), Hangzhou, China, 2023*.

SELECTIVE PATENT

1. **X. Mei**, D. Han, Z. Wu, et al. "A Target Localization Approach Based on Information Fusion in Marine Search and Rescue Wireless Sensor Networks", CN Patent, 202210536968.0. (Granted)
2. **X. Mei**, D. Han, Z. Wu, et al. "A Joint Estimation Method for Target Location and Environment Propagation Parameter in Underwater Wireless Sensor Networks", CN Patent, 202210536969.5. (Granted)
3. **X. Mei**, H. Wu, D. Han et al. An Efficient Particle Filtering Method for Node Localization in Ocean Sensor Networks, CN Patent 202310390266.0. (Granted)
4. **X. Mei**, D. Han, Z. Wu, et al. "An Optimal Path Planning Method of Autonomous Surface Vehicles for Target Localization Error Minimization", CN Patent, 202210536357.6. (Granted)
5. **X. Mei**, H. Wu, D. Han et al. Positioning Method of Water Surface Sensor Network with Unknown Signal Propagation Loss under Non-line-of-sight Condition, South Africa Patent, 2023/00699. (Granted)
6. **X. Mei**, H. Wu, J. Xian et al. Fault-tolerant Positioning Device for Underwater Wireless Sensor Networks with Layered Effect, Nigeria Patent, F/PT/C/2024/11395. (Granted)
7. **X. Mei**, F. Miao, H. Wu et al. A Quantum-Optimization-Framework-Driven Optimal Deployment Method for Localization in Ocean Sensor Networks, CN Patent, 202511142876.4. (Substantive examination)
8. **X. Mei**, S. Han, H. Wu et al.A Multi-Source Iterative Localization Method for Ocean Wireless Sensor Networks under Non-Line-of-Sight and Stratified Propagation Conditions, CN Patent, 202511493263.5. (Substantive examination)
9. **X. Mei**, F. Mian, H. Wu et al. A Surface Sensor Network Positioning Method Integrating Quantum Computing under Dual Parameter Uncertainty Conditions, CN Patent, 202410669682.9. (Substantive examination)
10. **X. Mei**, H. Wu, J. Xian et al. A Fault-tolerant Positioning Method for Underwater Wireless Sensor Networks with Layered Effects, CN Patent, 202410174887.X. (Substantive examination)
11. **X. Mei**, H. Wu, D. Han et al. A Positioning Method for Ocean Sensor Networks with Uncertain Parameters, CN Patent, 202211471174.7. (Substantive examination)

12. **X. Mei**, H. Wu, D. Han et al. A Surface Sensor Network Positioning Method with Unknown Signal Propagation Loss under Non-line-of-sight Conditions, CN Patent, 202211419052.3. (Substantive examination)
13. **X. Mei**, D. Han, Z. Wu et al. A Target Localization Method using Information Fusion in Marine Search and Rescue Wireless Sensor Networks, PCT/CN2023/084273. (Publication)
14. **X. Mei**, D. Han, Z. Wu et al. An Optimal Path Planning for Localization Error Minimization of Autonomous Surface Vehicles, PCT/CN2023/084623. (Publication)
15. **X. Mei**, D. Han, Z. Wu et al. Joint Target Location and Signal Propagation Parameters Estimation in Underwater Wireless Sensor Networks, PCT/CN2023/084119. (Publication)
16. **X. Mei**, H. Wu, D. Han et al. Efficient Particle Filter Node Positioning Method for Ocean Sensor Networks, American Patent, US 18/631,847. (Submitted)
17. **X. Mei**, F. Miao, H. Wu, et al. Positioning Method of Surface Wireless Sensor Networks Intergrating Quantum Computation under Condition of Two-Parameter Uncertainty, British Patent, GB2411891.1. (Submitted)
18. **X. Mei**, D. Han, J. Xian, et. al. "Ship-to-shore communication network management system V1.0", CN Software Patent, 2023SR0844369. (Granted)

HONOR AND AWARDS

- 2025 **Fourth Young Talent Support Program of the China Institute of Navigation**
  Awarded by the China Institute of Navigation
- 2025 **1st Prize of the Science and Technology Progress Award**
  Awarded by the China Institute of Navigation
- 2025 **Navigation Science and Technology Innovation Award (Team Award)**
  Awarded by the China Institute of Navigation
- 2025 **Annual Outstanding Paper Award**
  Awarded by the China Institute of Navigation
- 2025 **Conference Best Paper Award**
  Awarded by the China Institute of Navigation
- 2025 **Visiting Scholar Award**
  Awarded by the China Scholarship Council
- 2025 **Best Paper Award**
  Awarded by the AIRSA 2025 Organizing Committee
- 2025 **2nd Prize of the 14th China Marine Vehicle Design and Construction Context**
  Awarded by the China Society of Naval Architects and Marine Engineers
- 2024 **Chenguang Scholar**
  Awarded by the Shanghai Education Development Foundation and Shanghai Municipal Education Commission
- 2024 **Yinhang Program**
  Awarded by the Shanghai Maritime University
- 2024 **China Navigation Academic Annual Conference Outstanding Paper**

Awarded by the China Institution of Navigation

- 2024 **2nd Prize of the 13th China Marine Vehicle Design and Construction Context**
  Awarded by the Chinese Society of Naval Architects and Marine Engineers
- 2023 **Grand Prize of the 12th China Marine Vehicle Design and Construction Context**
  Awarded by the Chinese Society of Naval Architects and Marine Engineers
- 2023 **1st Prize of the 12th China Marine Vehicle Design and Construction Context**
  Awarded by the Chinese Society of Naval Architects and Marine Engineers
- 2022 **Shanghai Post-doctoral Excellence Program**
  Awarded by the Shanghai Human Resources and Social Security Bureau
- 2022 **1st Prize of Science and Technology Progress Award**
  Awarded by the Chinese Society of Naval Architects and Marine Engineers
- 2022 **2nd Prize of Excellent Paper in the 19th Annual Academic Conference**
  Awarded by Shanghai Communication Society
- 2022 **Honorary Award of the 2nd Shanghai Postdoctoral Innovation and Entrepreneurship Competition**
  Awarded by Shanghai Human Resources and Social Security Bureau
- 2020 **National Scholarships for PhD Students**
  Awarded by National Ministry of Education.
- 2019 **National Study Abroad Fund**
  Awarded by China Scholarship Council (CSC).
- 2018 **Principal Scholarship**
  Awarded by Shanghai Maritime University, Office of Graduate Studies.
- 2018 **Shanghai Outstanding Graduate**
  Awarded by Shanghai Municipal Education Commission.
- 2017 **National Scholarships for Master Students**
  Awarded by National Ministry of Education.
- 2017 **Outstanding Students**
  Awarded by Shanghai Maritime University.
- 2017 **Principal Scholarship**
  Awarded by Shanghai Maritime University, Office of Graduate Studies.
- 2016 **Principal Scholarship**
  Awarded by Shanghai Maritime University, Office of Graduate Studies.
- 2016 **Shanghai Outstanding Graduate**
  Awarded by Shanghai Municipal Education Commission.

PROFESSIONAL AFFILIATIONS AND SERVICES

- **IEEE Member**
- **IEEE Communications Society Member**
- **Editorial Board in Discover Oceans**

- **Editorial Board in Discover Sensors**
- **Youth Editorial Board in Journal of Ocean University of China**
- **Youth Editorial Board in Intelligent Marine Technology and Systems**
- **Youth Editorial Board in The Ocean Engineering**
- **Early Career Editorial Board in Oceans**
- **Youth Editorial Board in Journal of Measurement Science and Instrumentation**
- **Junior Editorial Board in Complex Engineering Systems**
- **Youth Editorial Board in Electronics and Signal Processing**
- **Youth Editorial Board in Radio Engineering**
- **Youth Editorial Board in Ocean Development and Management**
- **Review expert for the EU Marie Curie Actions co-funded project (EUTOPIA-SIF)**
- **OCOT' 2026 Technical Program Committee Member**
- **ICPST' 2026 Technical Program Committee Member**
- **PSGEC' 2026 Technical Program Committee Member**
- **ICMST' 2025 Technical Program Committee Member**
- **OTCN' 2025 Technical Program Committee Member**
- **NESP' 2025 Technical Program Committee Member**
- **ASGT' 2025 Technical Program Committee Member**
- **ICCCS' 2025 Academic Committee Member (IEEE)**
- **ICAIT' 2024 Publication Chair**
- **ICCCS' 2024 Academic Committee Member (IEEE)**
- **ICSGE' 2024 Technical Program Committee Member**
- **ICPRE' 2024 Technical Program Committee Member (IEEE)**
- **ICCCS' 2023 Technical Program Committee Member、Session Chair (IEEE)**
- **iCaMaL' 2023 Technical Program Committee Member (IEEE)**
- **ICPRE' 2023 Technical Program Committee Member (IEEE)**
- **NCWC'2023 Technical Program Committee Member**